

Micro I, class 22

I wrote down the solution for the linear-demand, identical-and-constant-returns technologies case. As the number of firms grows, the unique equilibrium price converges to the common unit cost, and the allocation of goods converges to a competitive allocation. Continuing with the Cournot model of oligopoly (identical products version), I applied the D-F-K theorem to a general identical-products Cournot game by specializing the inverse demand and cost functions of the firms. Previously,

I then introduced the Bertrand model of competition for the identical and constant-returns technologies case. We confirmed that the unique Nash equilibrium is that each firm charges the common unit cost (an equilibrium that is in weakly dominated strategies): with only *two* firms, a competitive allocation is realized. The conclusion, however, is fragile, as I mentioned: relaxing the identical products or constant returns assumptions can overturn the conclusion. As we will also see, repeated interaction can also overturn the conclusion.

Next up, I introduce *extensive form games*, which we will continue on Thursday. I have notes on it on Canvas. I will be following Osborne-Rubinstein's definition more closely than MWGs, for what it is worth.